

SETTING UP PC TO CONTROL POST-CURING PARAMETERS

The process is controlled and monitored through the Python code named “arduino_terminal.py” that is to be run using the desktop app Visual Studio Code (VSCode).

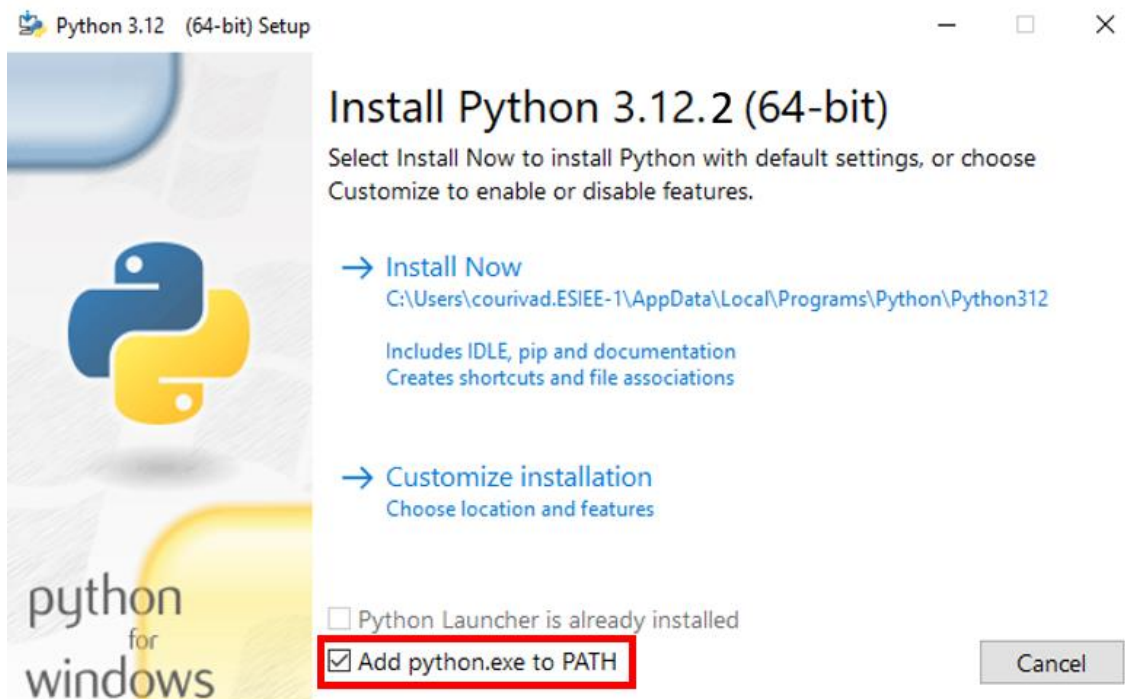
The Arduino board needs to be connected to the computer through the usb cord. It is useful to install the program Arduino IDE, but you don’t need to open or edit anything on it as the sketch “arduino_control_parameters.ino” has already been uploaded to the board.

If after reading this document you still have some doubts please check [Getting Started with Python in VS Code](#) as it is the backbone of this guide.

1. Install Python on the PC from Official Website

- a. install Python 3.12.2 (other versions might work as well, but they haven’t been tested)

Download the version from the official website and run the setup. Make sure to tick the “Add python.exe to PATH” option **before** clicking on “Install Now”.



- b. if Python is already installed on your PC but you’re unsure of which version you’re running then go to the Command Prompt window and insert the line “python -V” to check:

```
C:\Users\mabarb>python -V
Python 3.12.2
```

If this simple check does not work, something went wrong during the installation and you should fix it before proceeding with the next steps. It might be something related to the PATH and environmental variables of your computer.

If you’re using a different version, it is advised you uninstall Python and install this version if you encounter any issues with the script when running the code.

```
C:\Users\mabarb>pip --version
pip 25.1.1 from C:\Users\mabarb\AppData\Roaming\Python\Python312\site-packages\pip (python 3.12)
```

With the command “pip –version” check if pip has been properly installed together with Python (it should have been installed automatically when clicking on “Install Now”).

- c. After installing Python it is important to also install all the packages that are used in the script. To do so you need to insert some the following command on the Command Prompt window “pip install -name of the package to install-“.

```
C:\Users\mabarb>pip install pyserial
Defaulting to user installation because normal site-packages is not writeable
Collecting pyserial
  Downloading pyserial-3.5-py2.py3-none-any.whl.metadata (1.6 kB)
  Downloading pyserial-3.5-py2.py3-none-any.whl (90 kB)
    90.6/90.6 kB 854.7 kB/s eta 0:00:00
Installing collected packages: pyserial
  WARNING: The scripts pyserial-miniterm.exe and pyserial-ports.exe are installed in 'C:\Users\mabarb\AppData\Roaming\Python\Python312\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pyserial-3.5

[notice] A new release of pip is available: 24.0 -> 25.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\mabarb>python.exe -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in c:\program files\python312\lib\site-packages (24.0)
Collecting pip
  Downloading pip-25.1.1-py3-none-any.whl.metadata (3.6 kB)
  Downloading pip-25.1.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 6.1 MB/s eta 0:00:00
Installing collected packages: pip
  WARNING: The scripts pip.exe, pip3.12.exe and pip3.exe are installed in 'C:\Users\mabarb\AppData\Roaming\Python\Python312\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-25.1.1

C:\Users\mabarb>
```

You might get the message that a new release of pip is available, run “python.exe -m pip install –upgrade pip” and update to the latest pip version.

You can ignore the WARNINGS and proceed to install the following packages:

- pyserial
- datetime
- matplotlib
- numpy
- pandas

Some of those might come preinstalled with the Python version itself, you can still run the command “pip install ----” and nothing bad will happen if that package is already present on your PC.

You shouldn’t encounter any issues during this step, but if you do there are plenty of forums and tutorials on how to do this on the internet.

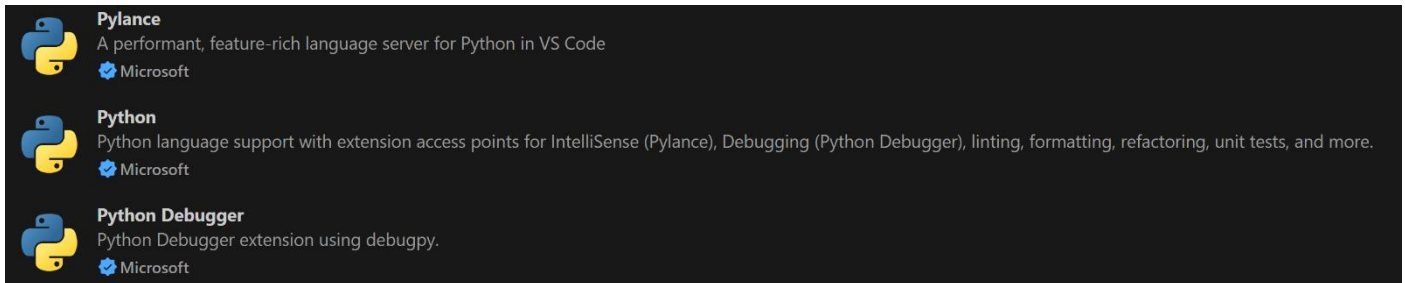
Don’t move onto the next steps unless you’re sure everything has been properly installed.

Once you’re done close the Command Prompt window

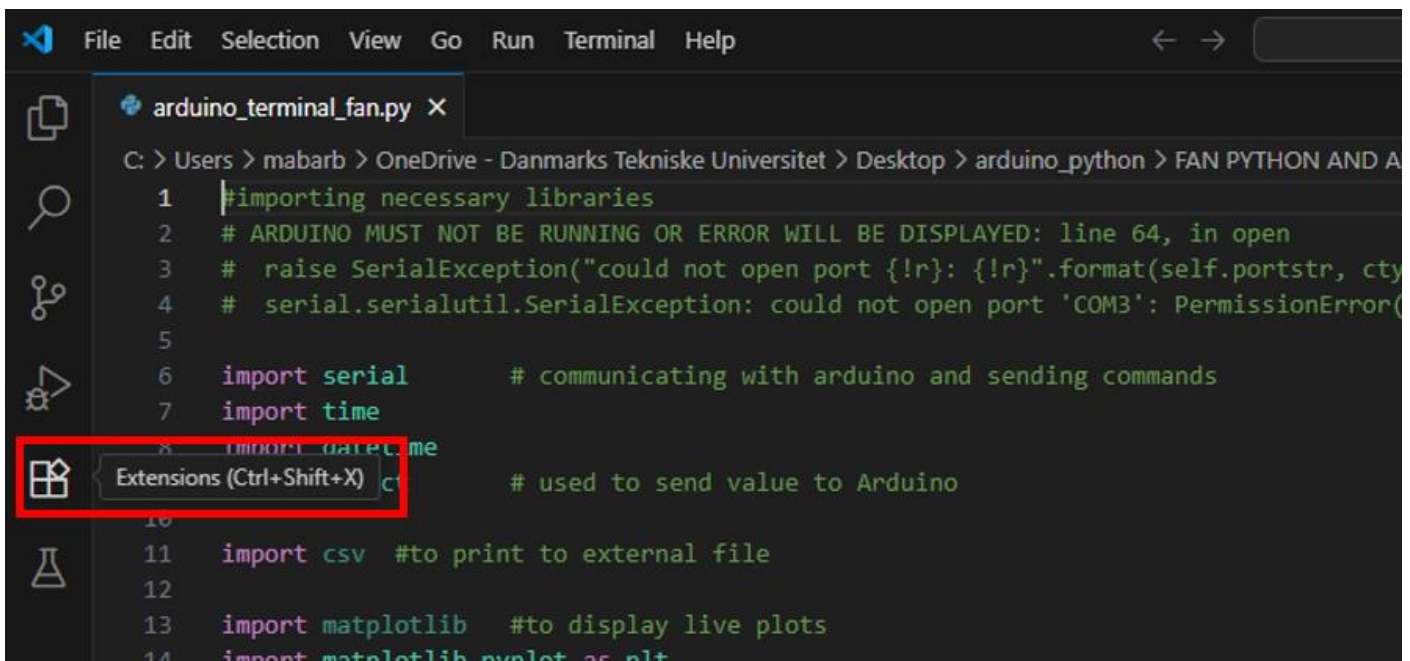
2. Install VSCode from the official website

Since you might need to edit the Python code, you should install Visual Studio Code (VSCode).

a. After installing the program it is necessary to also install the following extensions:



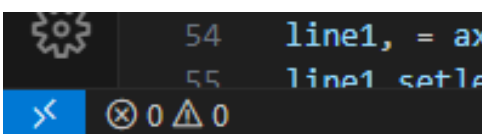
To do so you need to open the panel you see on the left side of the window



This will open a research bar to look for the extensions in the “Extensions Marketplace”

- Pylance
- Python
- Python Debugger

Try and simply open the python script (you need to download it and work with a local copy of it) with VSCode. Check the bottom left corner for any warnings, there should be none.

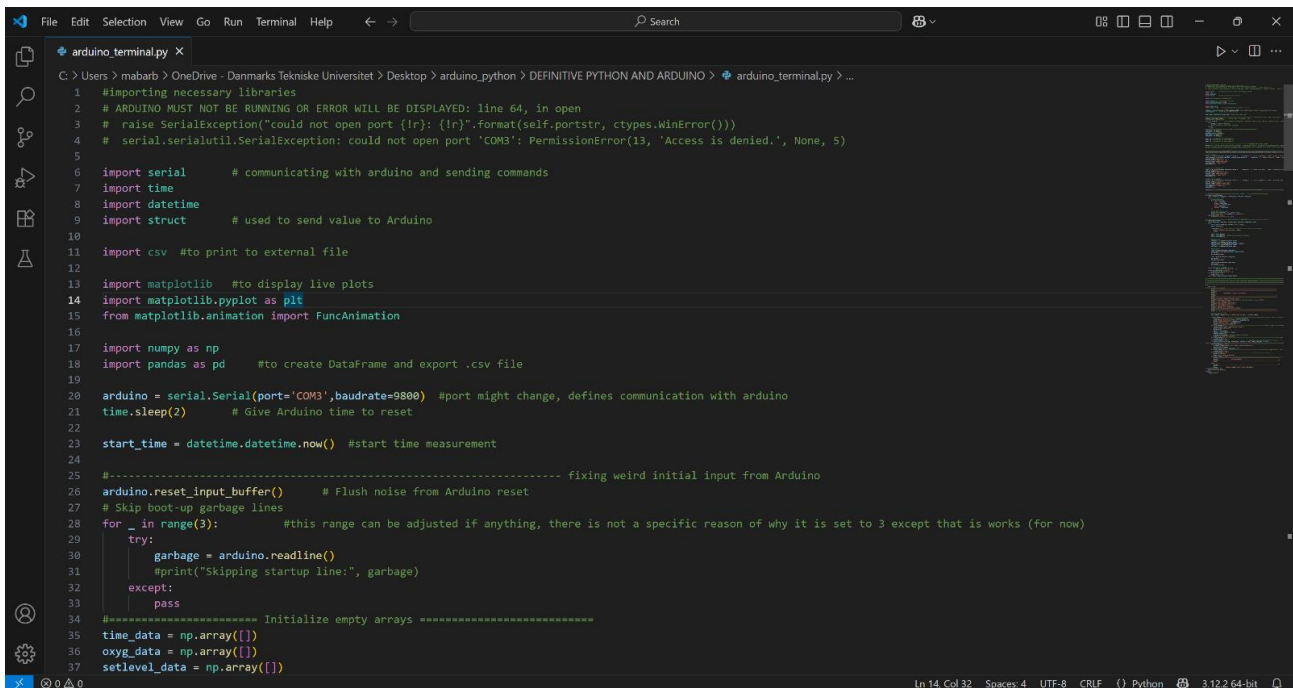


You might need to trust either the folder or the file (follow the instructions on VSCode to do that). Check that you've selected the right language in the bottom right corner of the window.

FINAL SETTING UP STEPS

In order to conduct the different experiments it is important to be able to run the various commands from the user interface from the terminal. Before doing this please ensure that all previous steps have been completed following the instructions in this guide (for example running the python code using the Spyder IDE compiler might lead to malfunctioning, so please avoid doing that).

Opening the “arduino_terminal.py” file using VSCode you should be able to see something like



```

1 #importing necessary libraries
2 # ARDUINO MUST NOT BE RUNNING OR ERROR WILL BE DISPLAYED: line 64, in open
3 # raise SerialException("could not open port {}: {}".format(self.portstr, ctypes.WinError()))
4 # serial.serialutil.SerialException: could not open port 'COM3': PermissionError(13, 'Access is denied.', None, 5)
5
6 import serial # communicating with arduino and sending commands
7 import time
8 import datetime
9 import struct # used to send value to Arduino
10
11 import csv #to print to external file
12
13 import matplotlib #to display live plots
14 import matplotlib.pyplot as plt
15 from matplotlib.animation import FuncAnimation
16
17 import numpy as np
18 import pandas as pd #to create DataFrame and export .csv file
19
20 arduino = serial.Serial(port='COM3',baudrate=9800) #port might change, defines communication with arduino
21 time.sleep(2) # Give Arduino time to reset
22
23 start_time = datetime.datetime.now() #start time measurement
24
25 #----- fixing weird initial input from Arduino
26 arduino.reset_input_buffer() # Flush noise from Arduino reset
27 # Skip boot-up garbage lines
28 for _ in range(3): #this range can be adjusted if anything, there is not a specific reason of why it is set to 3 except that it works (for now)
29     try:
30         garbage = arduino.readline()
31         print("Skipping startup line:", garbage)
32     except:
33         pass
34
35 #----- Initialize empty arrays -----
36 time_data = np.array([])
37 oxyg_data = np.array([])
38 setlevel_data = np.array([])
  
```

this:

You might need to edit line 20.

```
arduino = serial.Serial(port='COM3', baudrate = 9800)
```

To see which port works for your computer you need to launch the Arduino IDE.



On the blue bar close to the top left corner click on the “Arduino Uno” menu and check the port associated to the board (**you need to plug in the usb cable connected to the actual board**). In this case the correct port is “COM6” so, to have a functioning script, you need to change line 20 in the python file to:

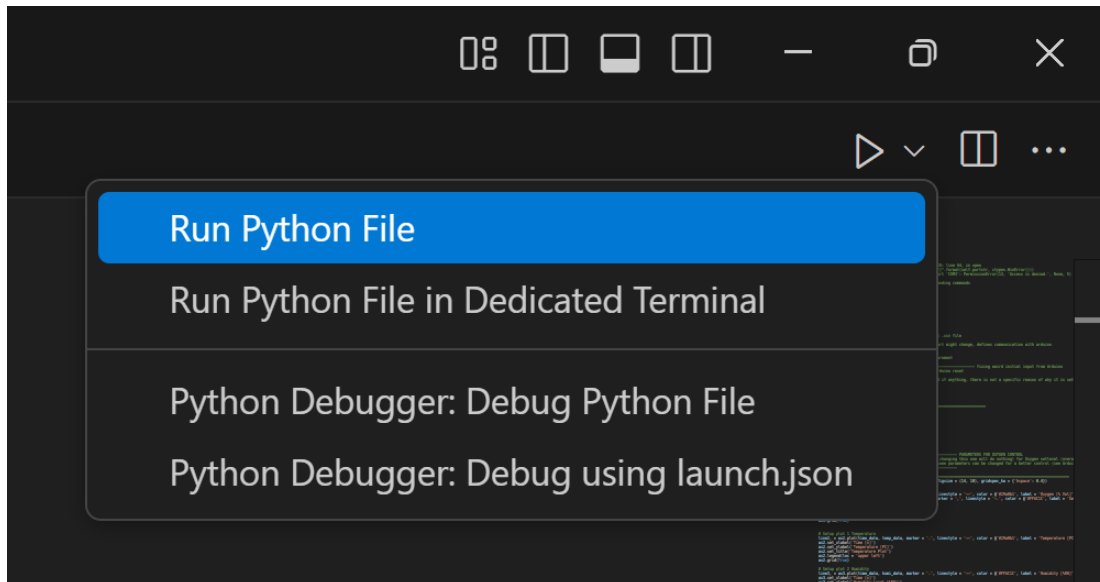
```
arduino = serial.Serial(port='COM6', baudrate = 9800)
```

Don't edit anything else from the script in VSCode, besides being an editor VSCode also allows you to run the code in the terminal with an interface similar to the one of MATLAB for those who might be familiar with it.

Close the Arduino IDE before going back to working on the python script.

RUNNING THE SCRIPT (this steps are necessary each time you want to run the script)

By clicking on the “play” icon in the top right corner of the screen you will open the terminal. You can also click to open the menu and select either the first or the second option.



Please be aware that you need to connect the board to the PC before running the Python file.

```
-----  
ENVIRONMENTAL CONTROL SETUP MENU:  
-----  
1: Start Fan  
2: Set Oxygen Setlevel [% Vol]  
3: Start Oxygen Control  
4: Start Animated Plot  
5: Reset Data Acquisition  
6: Display current ECS data  
7: Print Data to .CSV File  
8: Stop Oxygen Control  
9: Stop Fan  
10: Exit Environmental Control Setup  
-----  
Enter a command from the menu: █
```

If everything is working fine (it should) you'll see the user interface in the terminal:

The interface is pretty much self-explanatory but there are a few things you should remember to do each time you run the code:

REMEMBER TO PLUG IN THE POWER SOURCE OF THE CONTROL SYSTEM ONLY AFTER CONNECTING THE ARDUINO BOARD, BE AWARE THAT YOU'LL SEE NO WARNING IF THE POWER IS MISSING SO THINGS WILL SIMPLY NOT WORK.

**REMEMBER TO OPEN THE NITROGEN TANK SO THAT THE GAS WILL BE ABLE TO FLOW.
TO DO THAT SAFELY FOLLOW THE PROCEDURE DESCRIBED FURTHER DOWN THIS
GUIDE.**

DON'T PLUG IN THE POWER SOURCE WITHOUT HAVING FIRST CONNECTED THE ARDUINO BOARD TO THE PC AS THIS WILL CAUSE THE NITROGEN TO FLOW UNCONTROLLED.

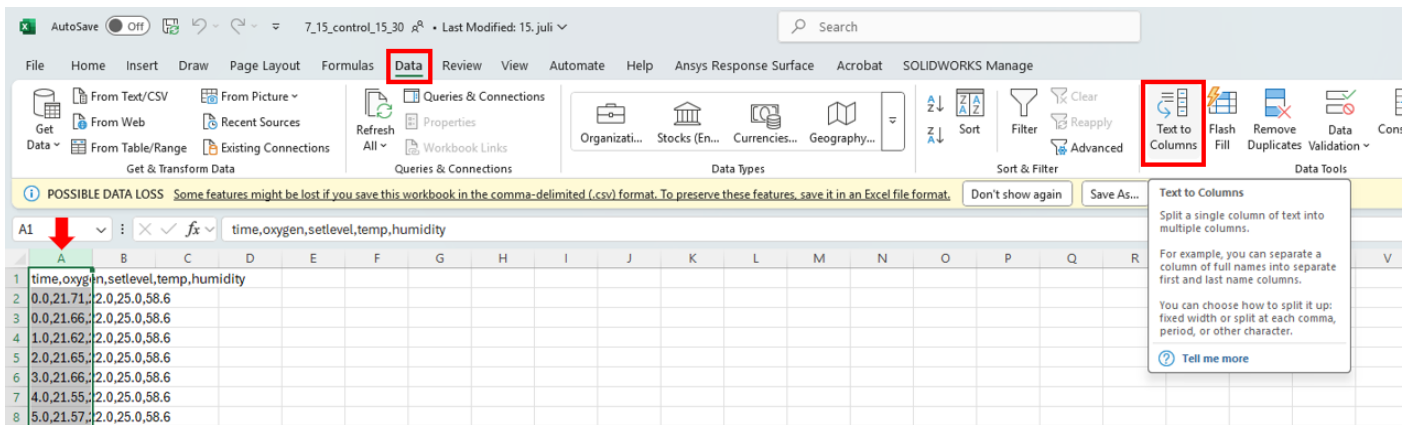
ONCE YOU'VE PLUGGED EVERYTHING IN AND OPENED THE TANK YOU CAN PROCEED. Everything is working properly if you can hear that the nitrogen is **not** flowing. Be aware that the power adapter sounds like a leak, everything is fine, the hiss of the nitrogen flowing is much louder than that.

- The first thing you should do is start the fan (pressing 1)
- The second thing you should do is set the oxygen Setlevel (pressing 2, then entering the numeric value and then pressing "enter")
- The third thing you should do is start the animated plot (pressing 4 then "enter")
- **REMEMBER TO START THE FAN**
- **REMEMBER TO OPEN THE NITROGEN VALVE AND START THE O₂ CONTROL** (pressing 3)
This will activate the control logic implemented in Arduino to regulate the oxygen content within the chamber. Please make sure that the nitrogen tank is unlocked and that the fan is on to ensure satisfactory oxygen control
- **Save the data to .csv file with a meaningful filename** (press 7, then follow the instructions in the terminal window (write filename))
- **BEFORE OPENING THE DOOR TO TAKE THE PARTS OUT STOP OXYGEN CONTROL** (press 8) You should also stop the fan but it's not that bad if it keeps running with the door open.
- Reset data acquisition before processing the next batch to restart the plot and data acquisition. (press 5) remember to first save the data in the .cvs output file otherwise it will be permanently lost

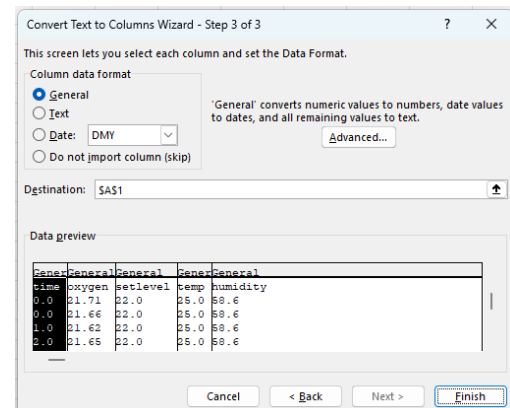
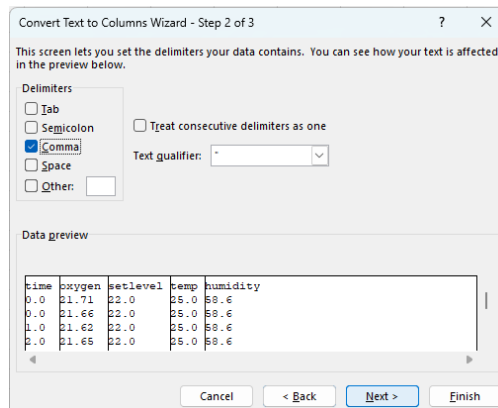
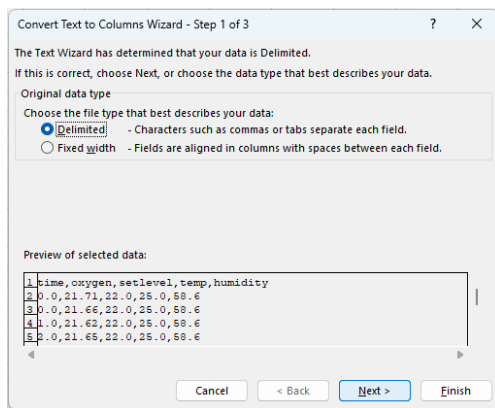
You can familiarize yourself with the script by running it with only the Arduino board connected (no power and nitrogen tank locked). You will be able to hear the fan rotating and the relays controlling the valve clicking as you start and stop the oxygen control (coherently with the oxygen setlevel value).

Try and test each of the commands to understand how they work.

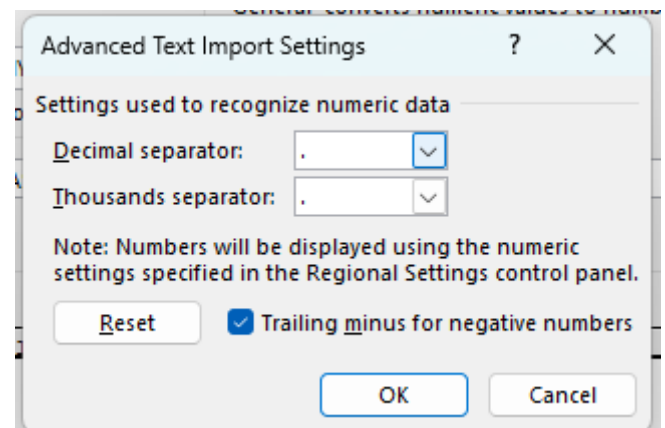
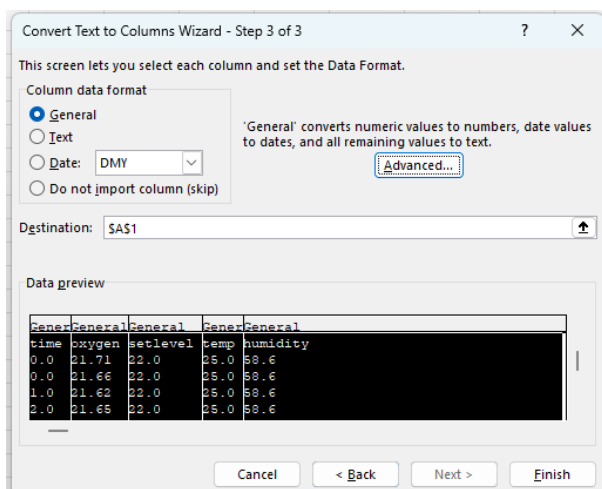
The ".csv" file saved is not comma separated so you need to edit it in excel to properly view the data collected.



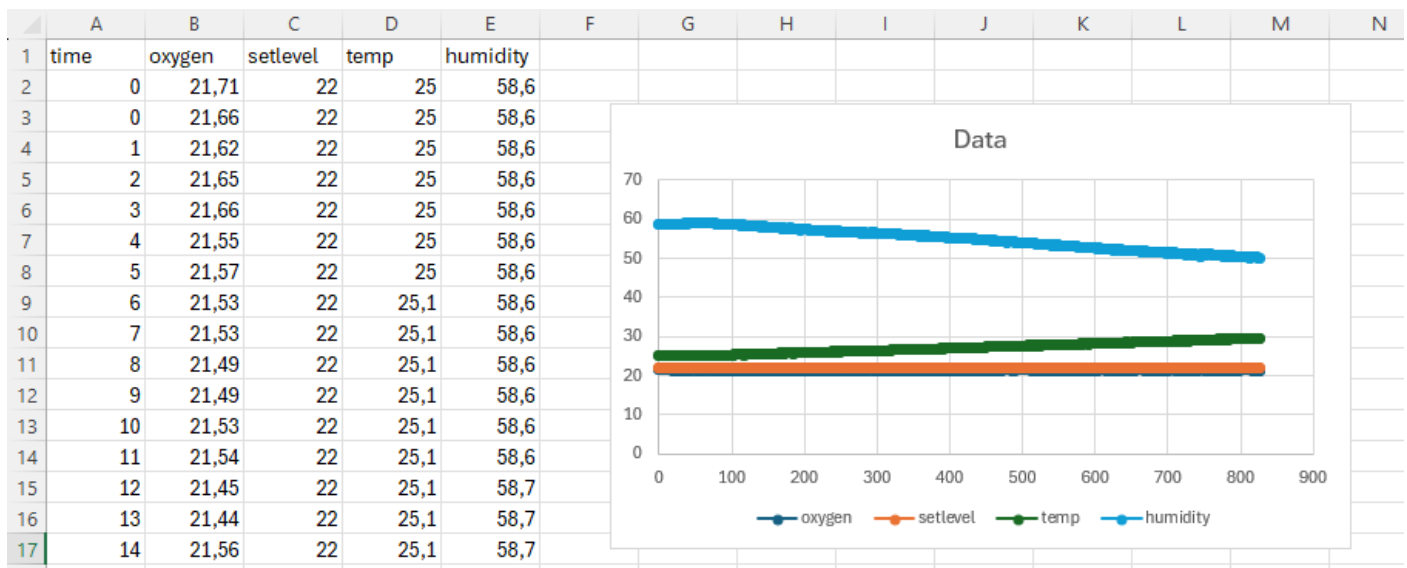
Click on “Data” in the upper bar and after selecting the first column select “Text to Columns” from the menu bar.



Once you get to the third step you need to click on the “Advanced” option after having selected the “General” option. Another pop-up will open; you need to change the “decimal separator” to a comma by typing it in. Leave everything else as is.



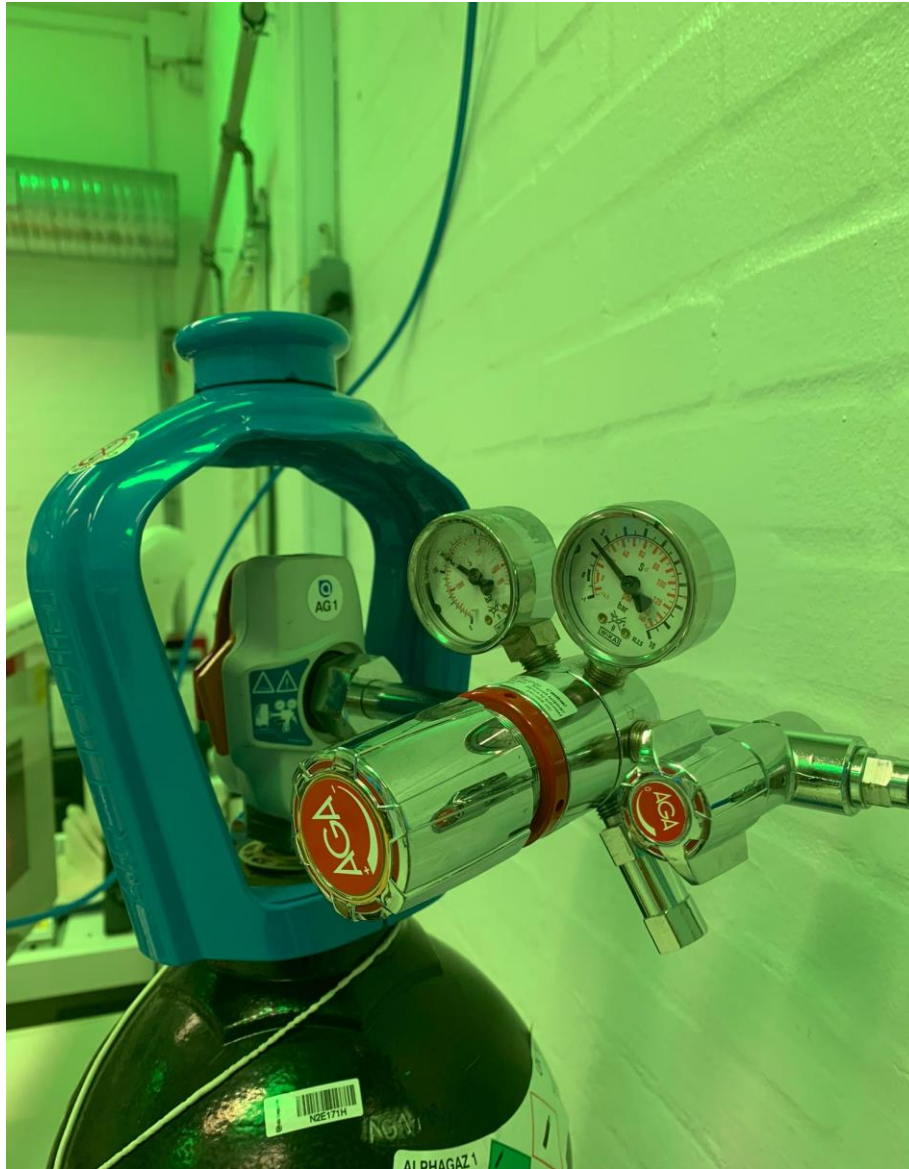
Click on OK and go back to the previous pop-up (step 3 of 3), hold down the shift key and left-click on each of the columns to select all of them. Once you’re done click on “Finish” .



You should be able to see and process the data in excel now.

HOW TO OPEN THE NITROGEN TANK

Worry about opening the tank only once the board has been plugged in the usb port and the power supply is connected. In these conditions no Nitrogen should flow even after opening the tank.



Nitrogen tank safely locked, no flow.

To safely close the tank, repeat the steps in the inverted order (closing first the knob and pushing down the levers (both of them at the same time, by pushing on the red one)).

**ALWAYS LOCK THE TANK ONCE YOU'RE DONE
USING IT**

Step 1:

Pull levers up, pull on the orange lever (this will consequently push the lever number 2 up as well)



Step 2:

Rotate knob counter-clockwise all the way, you should hear a short hiss. If the hiss keeps going, close back the knob rotating clockwise and understand where the nitrogen is coming from (probably the valve, check that the Arduino board is powered and connected to the computer) before letting the nitrogen flow.



Nitrogen tank open, flow allowed.

POST-CURING

Once everything else has been dealt with and double checked you can start post processing the printed parts. Make sure that everything is plugged in, including the post-curing machine.



On the back of the machine there is a switch to turn on the machine.

Once the machine is turned on you can open the door touching the sensor on the front of the machine, above the door. Since sometimes this does not work, you can use the mechanical button (use an Allen wrench to press it and unlock the door).

Place the parts you want to post-cure on the glass tray (only after having properly washed them with IPA).

Close back the door.

DO NOT START THE POST-CURING YET!



Look at the display below the door, using the knob at its right select "Ordinary curing" by pressing it down. Click on "Drying time" and by rotating the knob put it to 0 min. Similarly select and change the "UV curing time". **Select "Heating time" and put it to 0 min, just to be safe select "Heating temperature" and put it to 60°C. The sensors cannot operate at temperatures higher than that.**

BEFORE STARTING THE POST-CURING START THE OXYGEN CONTROL WITH THE DESIRED SETTINGS AND WAIT FOR THE OXYGEN CONTENT TO REACH AND MAINTAIN THE SETLEVEL. WHEN THE OXYGEN IS WHERE IT SHOULD BE YOU CAN START THE POST-CURING.